

Guidelines of B.Sc. (H) Computer Science VI Semester/ B.A. Programme VI Semester
DSC-A6/DSE: DEEP LEARNING
(NEP UGCF 2022)
(Effective from Academic Year 2024-25)

S.No	Topic	Chapter	Reference	No. of hours
1	Unit 1 Introduction to neural networks: Artificial neurons, perceptron, computational models of neurons, Structure of neural networks, Multilayer feedforward neural networks (MLFFNN), Backpropagation learning, Empirical risk minimization, bias-variance tradeoff, Regularization, output units: linear, softmax, hidden units: tanh, RELU	1 6 (6.1, 6.2, 6.3, 6.4, 6.5.1, 6.5.2, 6.5.4)	[2] [1]	(8 Hours)
2	Unit 2 Deep neural networks: Difficulty of training DNNs, Greedy layerwise training, Optimization for training DNN's, Newer optimization methods for neural networks (AdaGrad, RMSProp, Adam), Regularization methods (dropout, drop connect, batch normalization).	7 (7.1) 8 (8.2, 8.3.1, 8.3.2, 8.5.1, 8.5.2, 8.5.3, 8.7.1)	[1]	(8 Hours)
3	Unit 3 Convolution neural networks(CNNs): Introduction to CNN - convolution, pooling, Deep CNNs - LeNet, AlexNet. Training CNNs, weights initialization, batch normalization, hyperparameter optimization, Using a pre trained convnet	9 (9.1, 9.2, 9.3, 9.4)	[1]	(8 Hours)
4	Unit 4 Recurrent neural networks (RNNs): Sequence modeling using RNNs, Backpropagation through time, LongShort Term Memory (LSTM), Bidirectional RNN	10 (10.1, 10.2 upto pg 372, 10.3, 10.7, 10.10.1)	[1]	(8 Hours)
5	Unit 5 Unsupervised deep learning: Autoencoders, Generative Adversarial Networks.	14 (14.1, 14.2) 20 (20.20.4)	[1] [1]	(8 Hours)
6	Unit 6 Applications: Computer vision, Speech recognition and NLP.	12 (12.2, 12.3, 12.4.1-12.4.4)	[1]	(5 Hours)

Essential/recommended readings

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press Book, 2016.
2. Bunduma, N., Fundamentals of Deep Learning, 1st edition, O'reilly Books, 2017.
3. Francois Chollet, Deep Learning with python, 2nd edition, Meaning Publications Co, 2021.(Can be used for Practicals)

Additional References

1. 12. Heaton, J., Deep Learning and Neural Networks, 1st edition, Heaton Research Inc., 2015.

Suggested Practical List:

The following practical are to be conducted using Python.

1. Implement a deep-neural feed-forward network for estimating the price of house, given real-estate data (Boston Housing Price)
2. Implement a feed-forward neural networks for classifying movie reviews as positive or negative (using IMDB dataset)
3. Implement a deep-neural network for classifying news wires by topic (Reuters dataset).
4. Implement CNN for classifying MNIST dataset
5. Classify MNIST dataset using any pertained model
6. Create a model for time-series forecasting using RNN/LSTM
7. Implement an auto-encoder